



Web Programming

Template Engines
Lecture 19 & 20
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What is a Template Engine?

A template engine is a tool that helps us create **dynamic HTML pages** using data.

Instead of writing fixed (static) HTML, we can insert variables

```
<h1>Hello {{ name }}</h1>
```



Where is it Used?

Template engines are used in:

- Backend (server-side)
- Web frameworks like Express
- MVC architecture (View layer)

Flow: User → Server → Template Engine → HTML → Browser



How It Works

User requests a page

Server gets data (e.g., user name)

Template engine combines data + template

Final HTML is sent to browser

Popular Template Engines



- Nunjucks
- EJS
- Pug

Nunjunks

npm install express nunjucks

{% %} <!-- logic -->

{{ }} <!-- output -->

{# #} <!-- comment -->



File system

```
my-app/  
  app.js           ← Node/Express server  
  package.json  
  views/           ← templates (Nunjucks / EJS)  
  public/          ← CSS, JS, images
```

```
const express = require('express');
const nunjucks = require('nunjucks');

const app = express();

// configure nunjucks
nunjucks.configure('views', {
  autoescape: true,
  express: app
});

app.get('/', (req, res) => {
  res.render('index.html', { name: 'Ali' });
});

app.listen(3000, () => {
  console.log('Server running on http://localhost:3000');
});
```

autoescape: It automatically **protects your HTML from dangerous code (like scripts)**.

without template engine

```
app.get('/', (req, res) => {  
  const name = "Ali";  
  res.send(`<h1>Hello ${name}</h1>`);  
});
```


Express + Nunjucks Example



```
app.get('/', (req, res) => {  
  res.render('index.html', { name: 'Ali' });  
});
```

Template:

```
<h1>Hello {{ name }}</h1>
```

```
nunjucks.configure("views", {  
  express: app  
});
```



Nunjucks Basics

Nunjucks uses simple syntax:

Variables:

```
{{ name }}
```

Conditions:

```
{% if user %}  
  Hello {{ user }}  
{% endif %}
```



elseif

```
{% if user.role == "admin" %}  
<p>Welcome Admin</p>  
{% elif user.role == "teacher" %}  
<p>Welcome Teacher</p>  
{% elif user.role == "student" %}  
<p>Welcome Student</p>  
{% else %}  
<p>Welcome Guest</p>  
{% endif %}
```

Loops Example

```
{% for item in items %}  
  <li>{{ item }}</li>  
{% endfor %}
```



Filters

```
{{ variable | filterName }}
```

```
res.render("profile.njk", {  
  name: "mahhek",  
  bio: "<b>Teacher</b>"  
});
```

```
{{ name | capitalize }}
```

Teacher (shown as text)

```
{{ bio | safe }}
```

 (make it bold)



String filters

{{ name | upper }}

{{ name | lower }}

{{ name | capitalize }} (**Converts only the first letter of the entire sentence uppercase**)

{{ title | title }} (**Converts first letter of EACH word to uppercase**)

{{ text | trim }} Removes **extra spaces from start and end**

{{ text | truncate(10) }} **10 = max number of characters**

It cuts the text if it's longer

Adds . . . by default - truncate(10, false) wont

{{ text | replace("a", "b") }}



Array filters

```
res.render("page.njk", {  
  items: ["banana", "apple", "mango"]  
});
```

```
<p>Length: {{ items | length }}</p>
```

```
<!-- 3 -->
```

```
<p>Join: {{ items | join(", ") }}</p>
```

```
<!-- banana, apple, mango -->
```

```
<p>First: {{ items | first }}</p>
```

```
<!-- banana -->
```

```
<p>Last: {{ items | last }}</p>
```

```
<!-- mango -->
```

```
<p>Reverse: {{ items | reverse | join(", ") }}</p>
```

```
<!-- mango, apple, banana -->
```

```
<p>Sorted: {{ items | sort | join(", ") }}</p>
```

```
<!-- apple, banana, mango -->
```



Filters on numbers

```
res.render("page.njk", {  
  price: 123.456,  
  count: "10"  
});
```

```
<p>Round: {{ price | round }}</p>          <!-- 123 -->
```

```
<p>Round (2 decimal): {{ price | round(2) }}</p>  
<!-- 123.46 -->
```

```
<p>Integer: {{ price | int }}</p>          <!-- 123 -->
```

```
<p>Float: {{ count | float }}</p>         <!-- 10 -->
```

```
<p>Float (from string): {{ "12.75" | float }}</p>  
<!-- 12.75 -->
```


List of Users

```
res.render("users.njk", {  
  users: [{ name: "mahhek", role: "teacher" },  
    { name: "ali", role: "student" }, { name: "sara", role:  
    "admin" }]});
```

```
</> nunjucks  
  
{% for user in users %}  
  <p>{{ user.name }} - {{ user.role }}</p>  
{% endfor %}  
  
<!-- Output:  
mahhek - teacher  
ali - student  
sara - admin  
-->
```

Template Inheritance

```
<!-- base.njk -->
<html>
  <body>
    {% block content %}{% endblock %}
  </body>
</html>
```

```
<!-- home.njk -->
{% extends "base.njk" %}

{% block content %}
  <h1>Home Page</h1>
{% endblock %}
```



Extend expects a block include doesn't

```
<header>
  <h1>My Website</h1>
  <nav>
    <a href="/">Home</a>
    <a href="/about">About</a>
  </nav>
</header>
```

```
{% include "header.njk" %}
```

```
<p>Welcome to the Home Page</p>
```



Conditions

```
{% if age > 18 and isStudent %}  
  Eligible  
{% endif %}
```

and, or, not

```
{# This is a comment #}
```

set



```
res.render("cart.njk", {  
  price: 100,  
  quantity: 3,  
  user: {  
    firstName: "mahhek",  
    lastName: "khan"  
  }  
});
```

<p>Total: {{ price * quantity }}</p>

<p>Name: {{ user.firstName + " " + user.lastName }}</p>

```
{% set total = price * quantity %}
```

```
{% set fullName = user.firstName + " " + user.lastName %}
```

<p>Total: {{ total }}</p>

<p>Name: {{ fullName | title }}</p>

raw

```
{% raw %}  
<p>{{ user.name }}</p>  
{% endraw %}
```

<p>{{ user.name }}</p>



EJS Basics

EJS (Embedded JavaScript) is a template engine used with Express.

It helps us create **dynamic HTML using JavaScript inside HTML**

```
<h1>Hello <%= name %></h1>
```

```
npm install express ejs
```

```
<%# This is a comment %>
```

```
<% %> → logic
```

```
<%= %> → output
```

File system

```
project/
├── app.js                # Main server file
├── package.json
├── views/                # 🔥 All EJS files go here
│   ├── partials/        # reusable components
│   │   ├── header.ejs
│   │   └── footer.ejs
│   ├── pages/           # main pages
│   │   ├── home.ejs
│   │   ├── about.ejs
│   │   └── users.ejs
│   └── layouts/ (optional)
│       └── main.ejs
├── public/              # static files
│   ├── css/
│   ├── js/
│   └── images/
└── routes/ (optional)
    └── index.js
```




```
<%- include("../partials/header") %>
```

```
<h1>Home Page</h1>
```

```
<%- include("../partials/footer") %>
```

EJS Syntax

Variables:

```
</> ejs  
  
<%= name %>
```

Logic (if condition):

```
</> ejs  
  
<% if (user) { %>  
    <h1>Hello <%= user %></h1>  
<% } %>
```



Loop Example (EJS)

```
<ul>
```

```
<% users.forEach(user => { %>
```

```
<li><%= user %></li>
```

```
<% }) %>
```

```
</ul>
```



Express + EJS Example

```
app.set('view engine', 'ejs');  
app.get('/', (req, res) => {  
  res.render('index', { name: 'Ali' });  
});
```

index.ejs:

```
<h1>Hello <%= name %></h1>
```



Resources

<https://mozilla.github.io/nunjucks/templating.html>

<https://ejs.co/>